Revenue A/B Testing Report (t-test)

1. Objective

To determine whether the new marketing campaign (Test group) generates higher average revenue per user compared to the existing campaign (Control group).

2. Metric

Revenue per user (continuous variable).

3. Statistical Method

A Welch’s t-test was used to compare the mean revenue between Control and Test groups. This test is appropriate for continuous data and does not assume equal variances.

4. Results

Average Revenue (Control): ■101.05
Average Revenue (Test): ■135.79
t-statistic: -5.172
p-value: 2.3493e-07

5. Interpretation

Since the p-value is less than 0.05, the difference in average revenue between the Test and Control groups is statistically significant.

6. Business Recommendation

Roll out the Test marketing campaign, as it generates higher average revenue per user and demonstrates clear statistical significance.